Sexual Reproduction in Plants

Exam code: 0610 & 0970

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Insect pollinated flowers

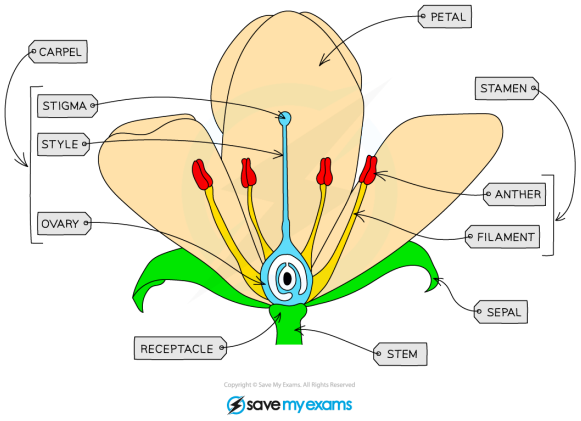
- Flowers are the **reproductive organs** of plants
- The role of flowers is to enable plant sex cells, or gametes to come together in **fertilisation**
  - The male gametes of plants are found in **pollen** grains
  - The female gametes of plants are in **ovules**
- The process by which pollen is transferred from the male part of a flower to the female part of a flower is known as **pollination**; this can be carried out in various ways, e.g. by insects or by wind

Insect pollinated flower structure

- The most well known flower structure is that of the **insect pollinated flower**
- The parts of an insect pollinated flower include:
  - sepals
  - petals
  - stamens
  - filaments
  - anthers
  - carpel
  - style
  - stigma
  - ovary
  - ovules

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand

Insect pollinated flower diagram



Insect pollinated flowers are adapted to attract insects and aid insect pollination

Flower structure and function table

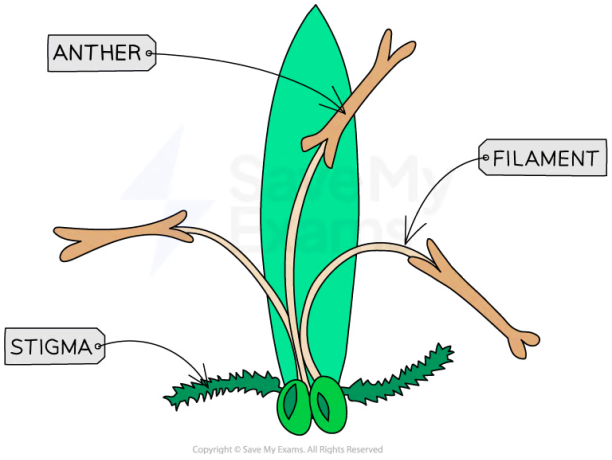
Structure	Description
Sepal	Protects unopened flower
Petals	Brightly coloured in insect-pollinated flowers to attract insects
Anther	Produces and releases pollen
Filaments	Provides support to the anther
Stigma	Sticky top of the female part of the flower which collects pollen grains
Style	A tube that connects the stigma and ovary
Ovary	Contains the ovules
Ovule	Structures inside the ovary that contain the female gametes

Did this video help you? Yes No

Wind pollinated flowers

- Wind pollinated flowers are adapted so that wind can easily catch pollen grains and carry them to the stigmas of other flowers
- The anthers and stigmas of wind pollinated flowers **hang outside the flower** so that:
  - pollen can easily be blown away by the wind
  - pollen can easily be caught by the stigmas of other flowers

Wind pollinated flower diagram



Wind pollinated flowers have anthers that hang outside the flower on long filaments, and feathery stigmas that can catch pollen easily

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Pollination & fertilisation

Pollen and pollination

- The pollen of different types of flower is adapted to aid pollination:
  - Insect-pollinated flowers produce **larger, heavier pollen grains** that often contain **spikes or hooks** on the outside so they are better able to stick to insects
  - Wind-pollinated flowers produce **small, lightweight pollen grains** that can be caught and carried easily by the wind

Worked Example

Observe the features of the different pollen grains

1. The pollen grains are large and have spikes around the outside
2. The pollen grains are small and smooth
3. The pollen grains are medium-sized and have wing-like structures on each side

Consider how each feature may relate to the pollination method

2. Small pollen grains will be light enough to be carried by the wind
3. Wing-like structures will catch the wind easily

Answer:

- **B** 1 only

Pollination in insect and wind pollinated flowers

- Insect and wind pollinated flowers have structural adaptations which aid pollination:

Structural adaptations of an insect pollinated flower table

Feature	Insect pollinated flower
Petals	Large and brightly coloured to attract insects
Scent and nectar	Scent and nectar are produced to encourage insects to visit the flower and push past stamen to get to nectar
Anthers	Held on stiff filaments within the flower so that they brush against insects
Stigma	Sticky stigmas within the flowers catch pollen grains when insects brush past

Structural adaptations of a wind pollinated flower table

Feature	Wind pollinated flower
Petals	Small and dull, often green or brown in colour
Scent and nectar	Scent and nectar are not produced; this would be a waste of energy
Anthers	Held on long filaments outside the flower to release pollen grains easily into the wind
Stigma	Feathery stigmas outside the flower catch airborne pollen grains

Pollination and fertilisation

- Pollination can be defined as:

The transfer of pollen grains from an anther to a stigma

- After pollination, the pollen nuclei travel to the ovule and fuse with female gametes in a process called fertilisation

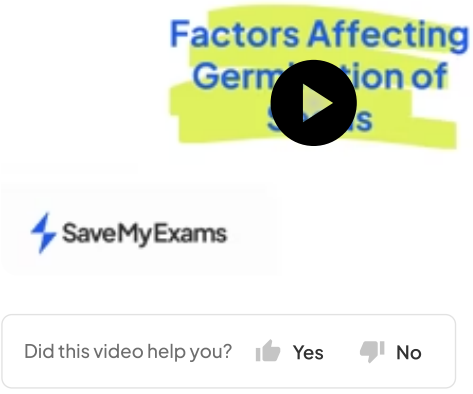
Plant fertilisation diagram

Pollen nuclei travel down a pollen tube in order to fuse with the female gametes inside the ovule; this is fertilisation

Examiner Tips and Tricks

Students often get confused between pollination and fertilisation in plants; remember that they are not the same thing!

- Pollination is the **transfer of pollen**
- Fertilisation is the **fusion of gametes**



Factors Affecting Germination of Seeds

in the embryo to start working so that growth can occur
    - **Oxygen** - so that energy can be released for germination
    - **Warmth** - germination improves as temperature rises (up to a maximum) as the reactions which take place are controlled by enzymes

- As carbon dioxide is not necessary for germination but also does not inhibit it, it makes no difference whether it is present or not

Investigating Germination

- Set up 4 boiling tubes each containing 10 cress seeds on cotton wool
- Set each test tube as shown in diagram below
- Leave tubes in set environment for a period of time: A, B and C incubated at 20°C; D placed in a fridge at 4°C
- Compare results and see which tube has the greatest number of germinated seeds

An investigation into the conditions required for germination of seedlings

Conditions required for germination results table

Test tube	Factor being tested	Seeds germinated
A	Water / moisture	No
B	Control (all factors present)	Yes
C	Oxygen	No
D	Warm temperature	No

- The results of this experiment prove that **all three factors are required** for the successful germination of the seeds and if any one factor is missing the seeds cannot germinate

Extended

- Cross-pollination occurs when the **pollen from one plant** is transferred to the stigma of **another plant of the same species**
- This is the way most plants carry out pollination as it **improves genetic variation**
- Occasionally, the pollen from a flower can **land on its own stigma** or on the **stigma of another flower on the same plant** - this is known as self-pollination
- Self-pollination **reduces genetic variety of the offspring** as all the gametes come from the same parent (and are therefore genetically identical)
- Lack of variation in the offspring is a disadvantage if environmental conditions change, as it is **less likely that any offspring will have adaptations that suit the new conditions** well
- On the other hand, cross-pollination relies completely on the presence of **pollinators** and this can be a problem if those pollinators are **missing** (e.g. the reduction in **bee** numbers is of great importance to humans as bees pollinate a large number of food crops) - this doesn't apply to wind-pollinated plants

Pollen tube formation

- After pollination, the following sequence of events take place:
  - A **pollen tube** grows from the pollen grain down the style until it reaches the ovary
  - **Nuclei from pollen** travel down the tube
  - The nuclei enter the ovule and fuse with the nucleus of the female gamete (fertilisation)
  - After fertilisation the **ovules develop into seeds and the ovary develops into a fruit**

Growth of the pollen tube and its entry into the ovule followed by fertilisation

Examiner Tips and Tricks

Remember that it is the **nucleus** of the pollen grain that travels down the pollen tube in the style for fertilisation and not the pollen grain itself. Make sure to be specific with your use of key words!

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